

A Deep Learning Approach for Enhancing Low-Resolution PET Brain Scans

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Preliminaries

Abstract

- PET is a functional imaging modality used to visualize biochemical processes in the body
- Ultra-high-performance dedicated brain PET scanners (such as NeuroExplorer) offer substantially higher resolution than conventional PET scanners
- However, their smaller detector elements and increased readout channels substantially raise cost, limiting availability
- Deep learning models trained on images from the dedicated high resolution brain scanners can enhance images from conventional lower resolution PET scanners at no added hardware cost
- Eight models were trained and tested on a paired dataset from the low resolution HRRT and high resolution NeuroExplorer brain PET scanners
- The best performing model was then evaluated on an independent dataset from a separate study using Siemens whole body and Radialis organ-targeted PET scanners

Objectives

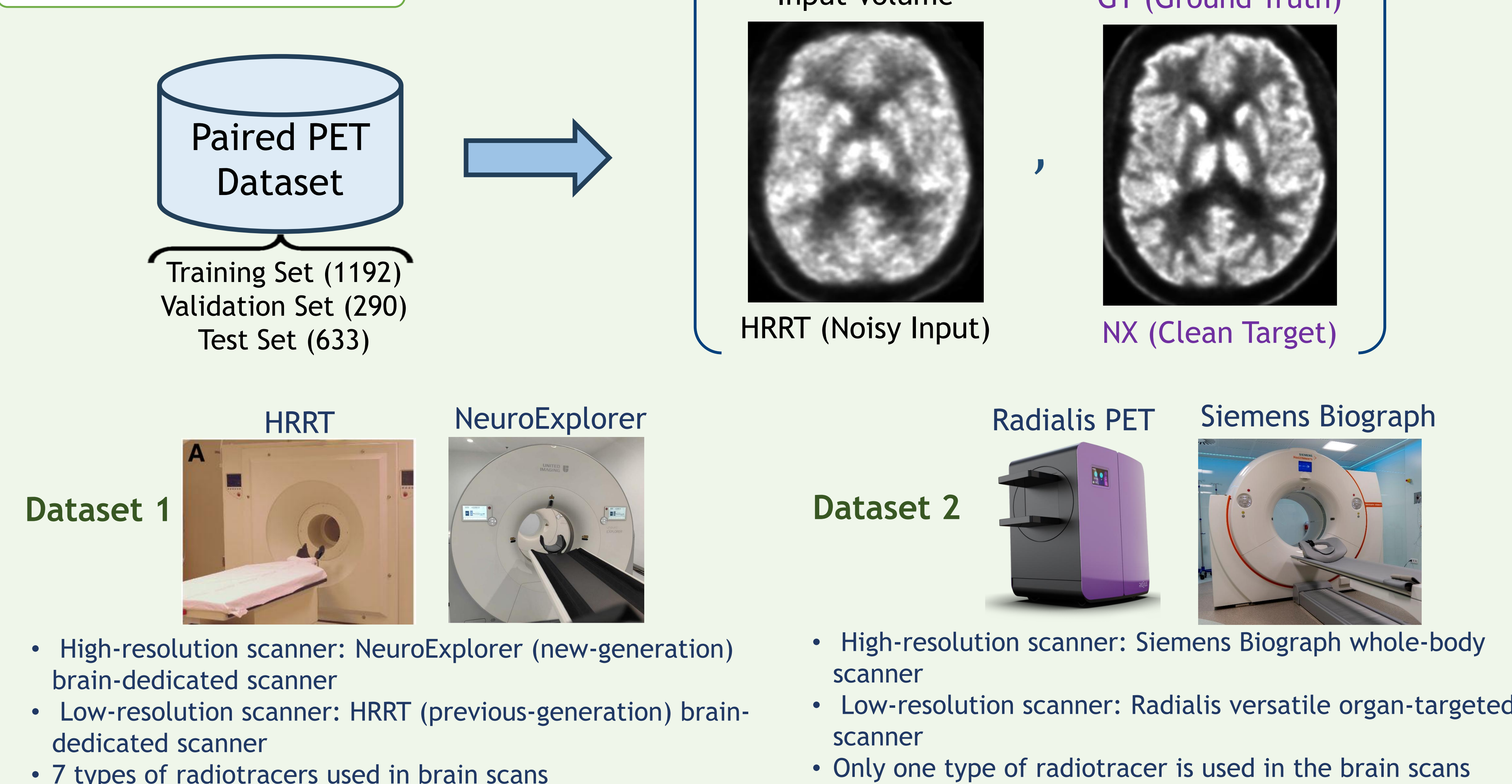
- Develop a deep learning model to convert low resolution into high resolution PET images
- Validate the model on paired datasets from two separate brain PET studies with different PET scanners

Challenges

- Small training dataset: only 7 subjects, each with a different radiotracer
- Accurate spatial alignment (registration) of paired PET images

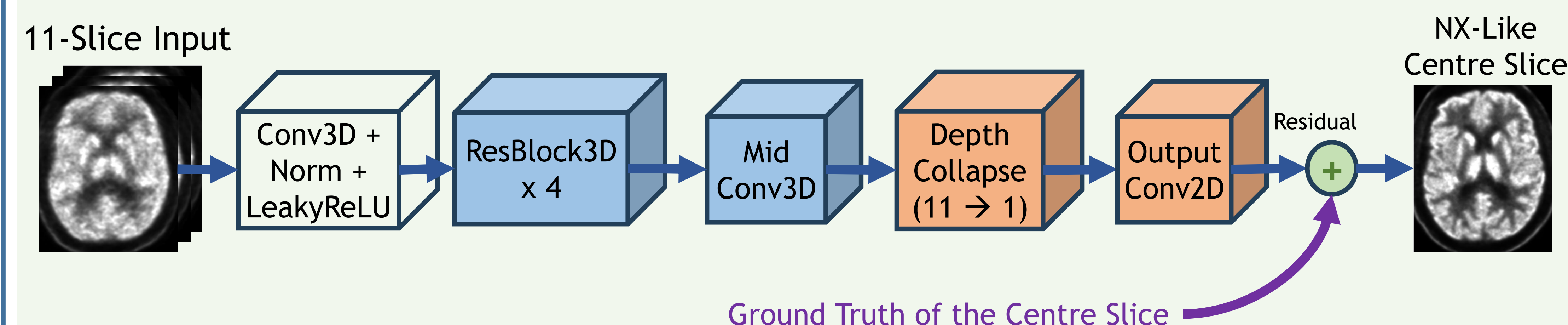
Proposed Solution

Collecting Datasets



Methodology

Architecture of the Best Performing Model



Dataset Curation:

- Paired HRRT and NX PET slices from 7 subjects were collected from publicly available datasets [1]
- Data were split by subject into training, validation, and test sets

Training Setup:

- Images were first screened for corrupted slices
- All slices were normalized to a 0-1 intensity range
- Sliding window inputs of 3, 5, 11, and 23 neighbouring slices were created

Evaluation Setup

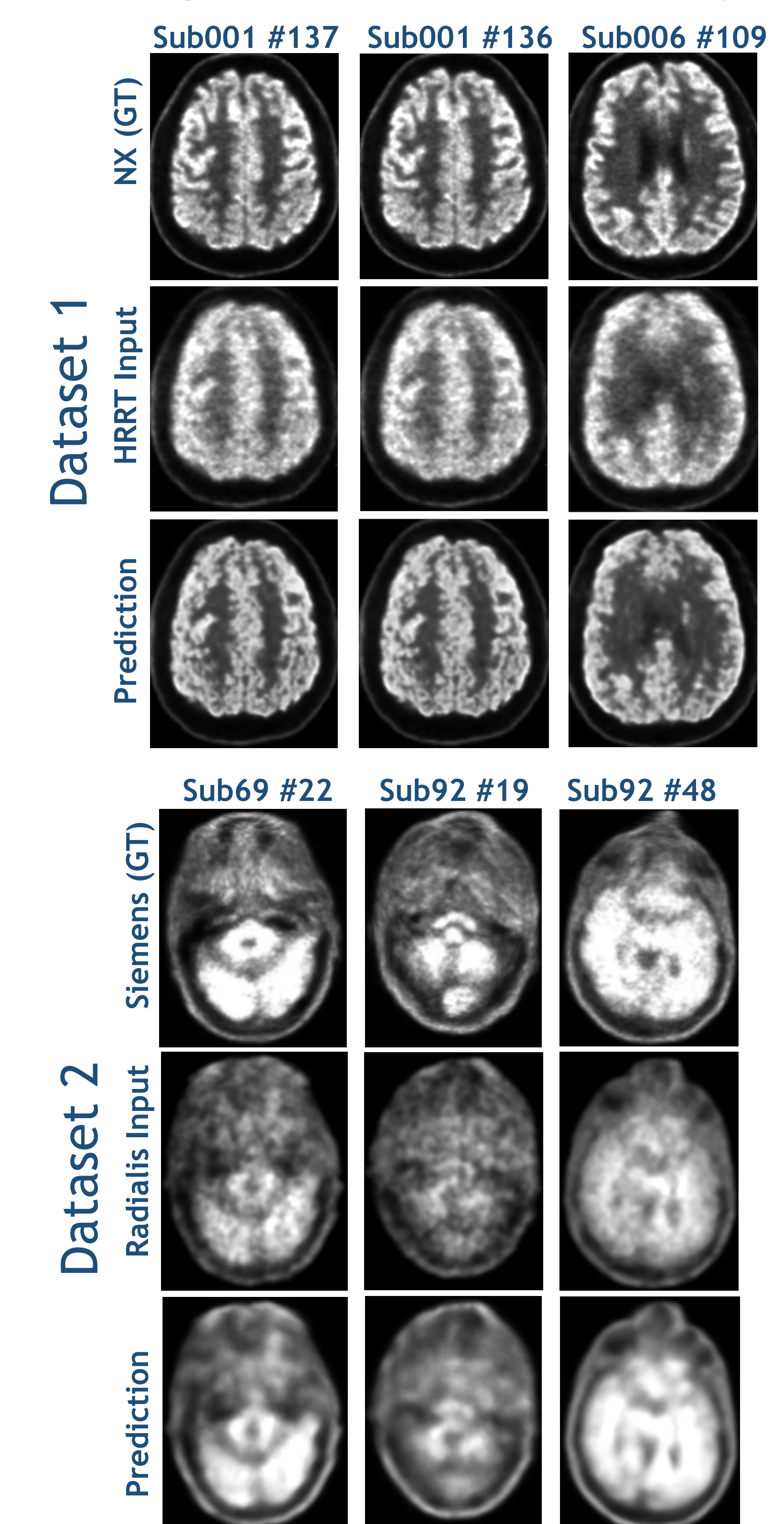
- Eight 3D models were trained with the same loss, optimizer, and split; the best four were retrained at each window size (16 setups)
- Results were compared using PSNR, SSIM, and MSE between the model output and the original HRRT baseline
- The best model was tested on data from the Radialis PET scanner (lower resolution) and Siemens Biograph scanner (higher resolution) pair supplied by research collaborators from an ongoing study consisting of 38 subjects

Experimental Results

Quantitative Analysis

Method	PET Image Score			PSNR Gain (dB)
	PSNR	SSIM	MSE	
Data Set 1: HRRT → NX				
Residual 3D CNN	21.64	0.6573	0.0072	0.9340 ↑
Data Set 2: Radialis PET → Siemens Biograph				
Residual 3D CNN	21.89	0.7117	0.0083	1.3559 ↑

Samples from Qualitative Analysis



References

- [1] T. Volpi et al., "Exceptional brain PET images from the NeuroEXPLORER: Scans with targeted radiopharmaceuticals and comparison to HRRT," *Eur. J. Nucl. Med. Mol. Imag.*, vol. 53, pp. 2110-2121, 2026, doi: 10.1007/s00259-025-07605-4.
- [2] T.-A. Song, S. R. Chowdhury, F. Yang, and J. Dutta, "Super-resolution PET imaging using convolutional neural networks," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 518-528, 2020, doi: 10.1109/TCI.2020.2964229.

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